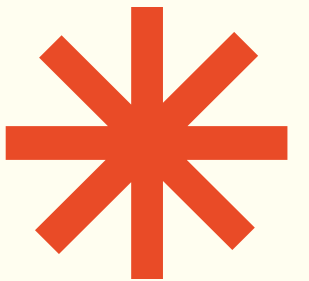


Optimizing Clinical Operations Through Enterprise Business Analytics at Aquila Health Network



Karen Lin

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Sponsor: Dan Stone | dds355@nyu.edu | Damark Group, Inc.

2026

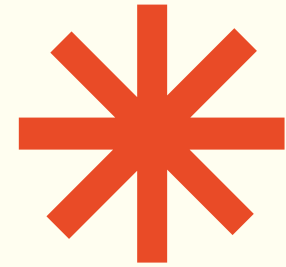
Applied Technical Project

Presentation in fulfillment of requirements of the NYU SPS course

Applied Technical Project: Capstone | MASY GC-4100

Spring 2026

Project Sponsor and Project Goal



Project Name: Optimizing Clinical Operations Through Enterprise Business Analytics at Aquila Health Network

Client Organization: Aquila Health Network (via Damark Group, Inc.)

Project Sponsor: Dan Stone, Damark Group President and NYU Adjunct Professor

Project Goal:

Establish a centralized business analytics platform to enable Aquila Health Network leadership to monitor clinical operations, optimize appointment utilization, and proactively manage patient access risk across 40 outpatient clinics in the Northeast.





What did the project needed to achieve?

Objective 01

Define Business and Functional Requirements (Feb 9 – Feb 23)

Deliver a validated FRS documenting KPIs, data requirements, and dashboard scope. Metric: Sponsor-approved requirements document.

Objective 02

Design Logical and Physical Data Models (Feb 24 – Mar 23)

Design star schema ERD and physical schema. Metric: Completed and validated data model aligned with reporting requirements.

Objective 03

Develop Analytics Dashboards (Mar 24 – Apr 13)

Build 5 interactive Tableau dashboards covering clinical operations, provider utilization, referral performance, and no-show risk. Metric: Fully functional dashboards with KPIs and filters.

Objective 04

Implement Predictive Analytics POC (Apr 14 – Apr 27)

Build logistic regression model to predict no-show risk. Metric: Working model with documented assumptions and insights.





What needed to be done?

1. Requirements Analysis

Documented business objectives, KPIs, and functional requirements through stakeholder collaboration with Dan Stone and Damark Group. Delivered FRS, Project Charter, WBS, and Gantt Chart.

2. Data Modeling

Designed a star schema data mart with FactAppointment as the central fact table and five dimension tables: DimPatient, DimProvider, DimClinic, DimDate, and DimReferral. Generated synthetic dataset representing 5,000 appointments across 40 clinics and 200 providers.

3. Dashboard Development

Built 5 interactive Tableau dashboards using the star schema data mart. Dashboards include cross-chart filtering, drill-through navigation, and filters for clinic, specialty, appointment type, and year.

4. Predictive Modeling

Developed logistic regression model using Python and scikit-learn with 7 features: prior no-show history, lead time, appointment hour, day of week, appointment type, insurance type, and specialty. Model documented in Jupyter notebook on GitHub.



Results

Data Mart

Star schema with 5 tables:
FactAppointment (5,000 records),
DimPatient (3,000), DimProvider (200),
DimClinic (40), FactReferrals (1,500),
and FactNoshowRisk (935 scored
appointments).

5 Interactive Tableau Dashboards

Clinical Operations | Revenue and
Access | Provider Utilization | Referral
Performance | No-Show Risk. All
dashboards include cross-filtering, drill-
through navigation, and
clinic/specialty/date filters.

Predictive Analytics Model

Logistic regression model (ROC-AUC:
0.595) identifying prior no-show
history as the dominant risk factor.
Model scores each upcoming
appointment as High, Medium, or Low
risk. Documented in Jupyter notebook
on GitHub.



Dashboard Results: Clinical Operations and Revenue

Clinical Operations Dashboard (Executive Summary)

Network KPIs: 5,000 total appointments | 18.1% no-show rate | 9.3% cancellation rate | 23.79 days average wait time | \$672,800 total revenue. Includes appointment status pie chart and heat map showing peak scheduling hours 11-14 on Tuesday through Thursday

Revenue and Access Dashboard

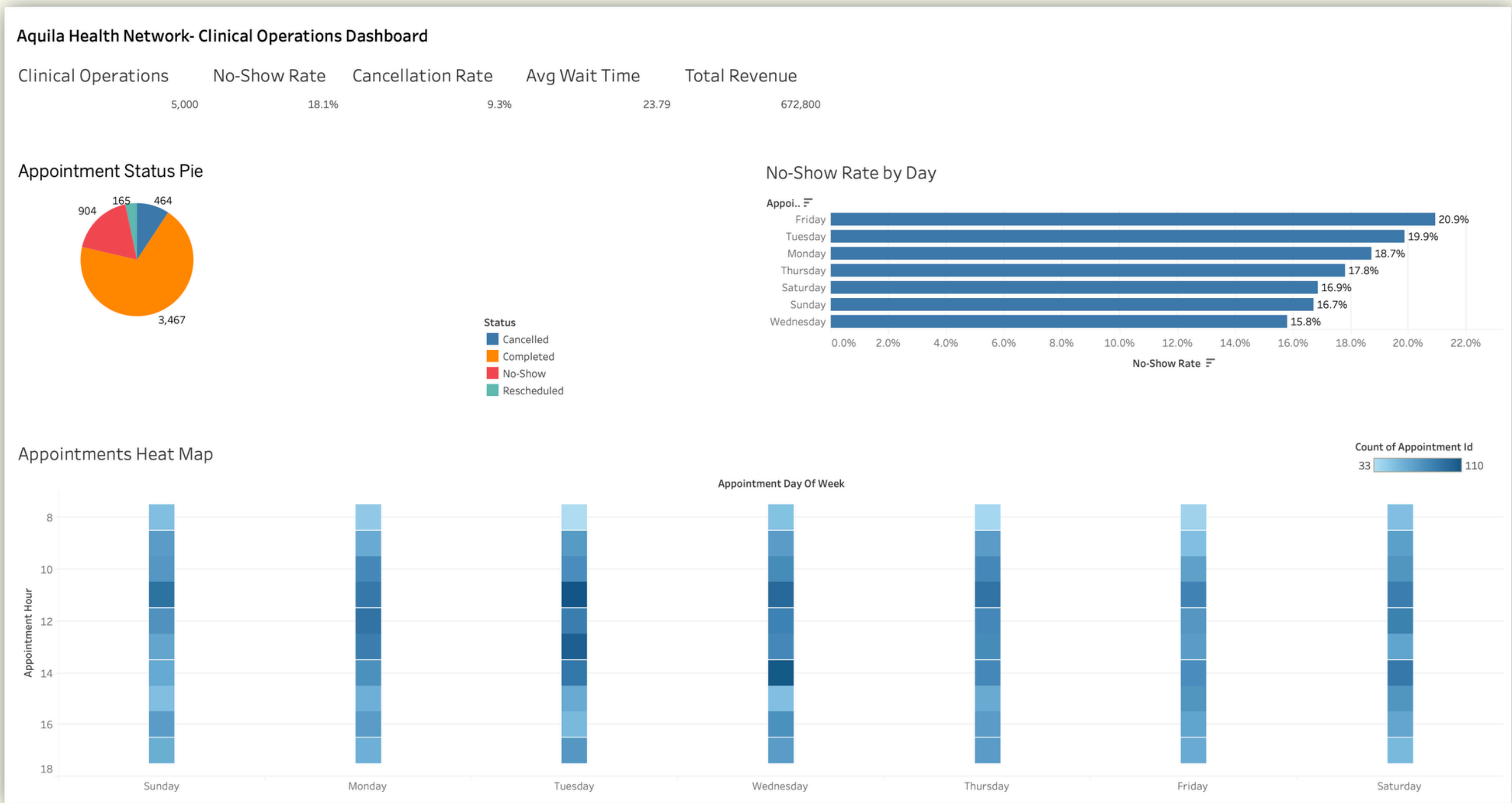
Revenue by Specialty: Urgent Care leads at \$113,285 | Pediatrics lowest at \$71,130. No-Show Rate by Specialty: Primary Care highest at 19.0% | Pediatrics lowest at 17.1%. Wait Time by Clinic: Paterson Community Clinic longest at 26.15 days | network average 23.79 days

Key Finding

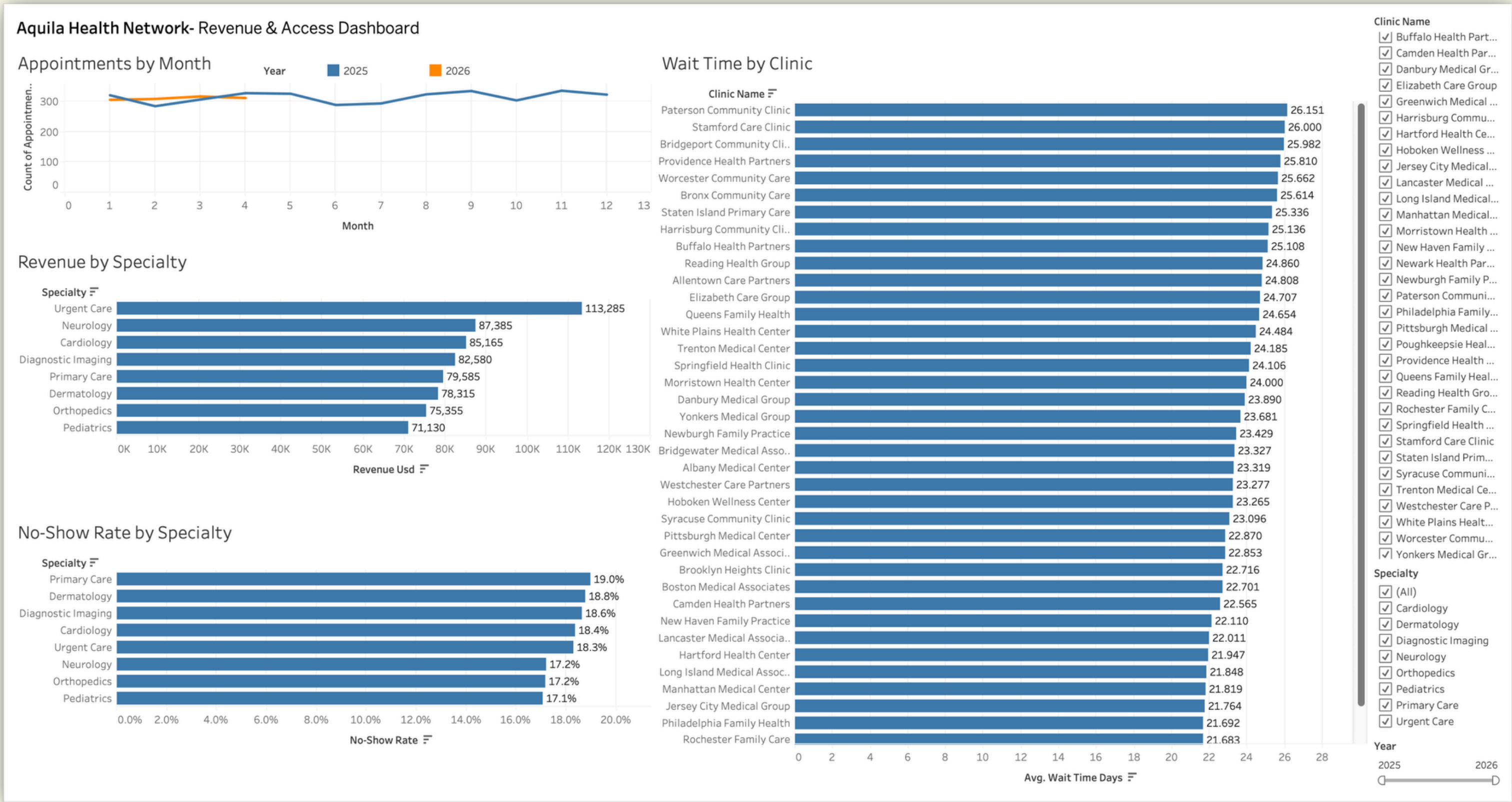
Friday appointments carry the highest no-show risk at 20.9% while Wednesday is lowest at 15.8% – directly actionable for scheduling and outreach prioritization.



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Dashboard Results: Provider Utilization and Referral Performance

Provider Utilization Dashboard

All 200 providers and 8 specialties plotted against the 82% utilization target (red dashed reference line). Key Finding: No specialty meets the 82% target. Rates range from 51.3% (Pediatrics) to 58.6% (Cardiology). Top providers: Dr. Elizabeth Clark DO and Dr. Kimberly Miller PA exceed target. Majority of providers fall significantly below.

Referral Performance Dashboard

Out of 1,500 total referrals: 704 Completed (47%) | 301 Scheduled | 201 Pending | 183 Not Scheduled | 111 Cancelled. Key Finding: 384 referrals (26%) are incomplete – representing revenue leakage and patient access gaps. Pattern is consistent across all 8 specialties.

Operational Implication

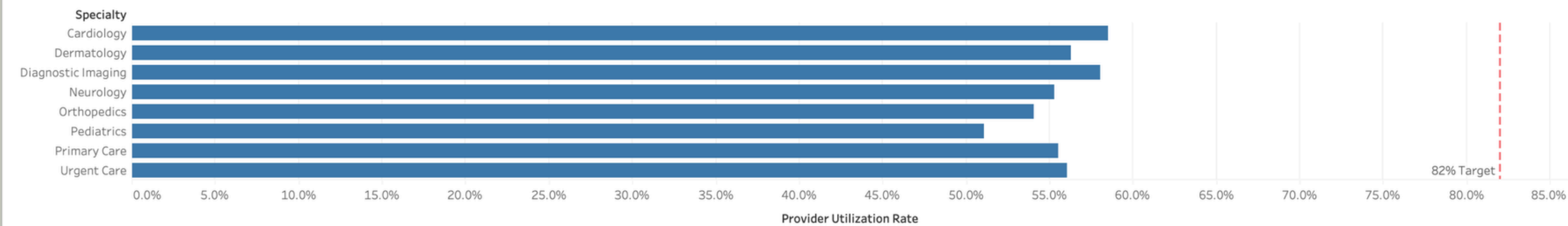
Utilization gap indicates scheduling inefficiency driven by no-shows and poor slot recovery. Referral incompleteness points to a need for automated follow-up workflows to capture lost revenue.



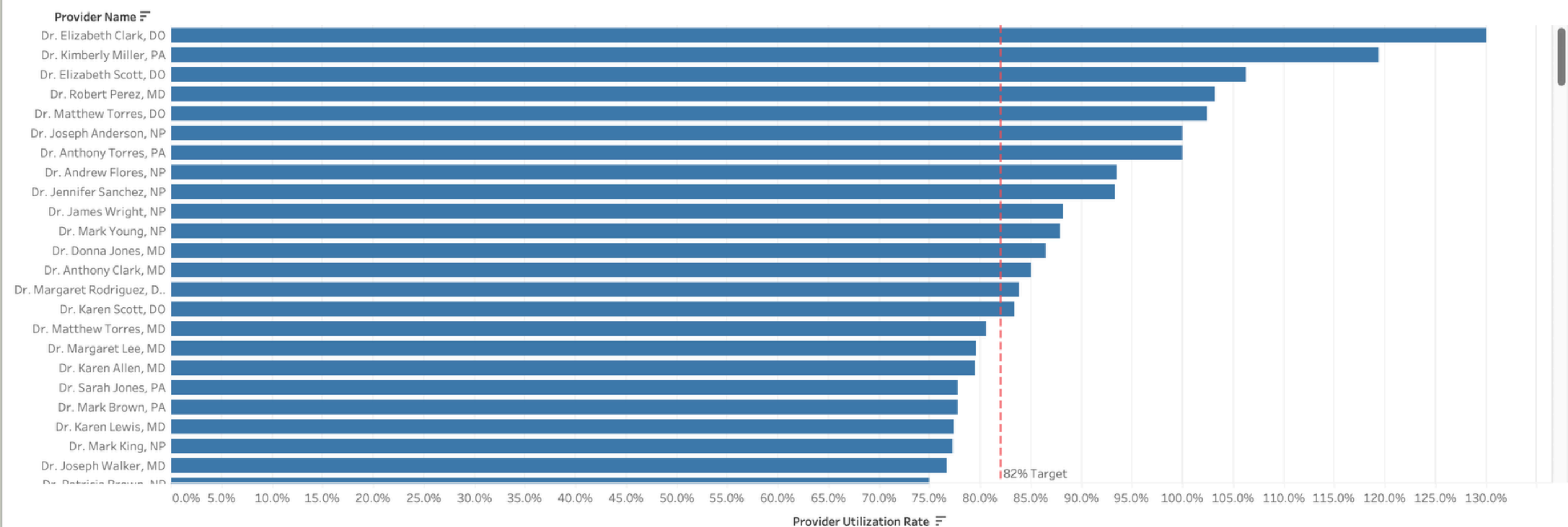
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Aquila Health Network- Provider Utilization Dashboard

Utilization by Specialty



Provider Utilization Rate



Clinic Name

- ☒ (All)
- ☒ Albany Medical Cen...
- ☒ Allentown Care Par...
- ☒ Boston Medical Ass...
- ☒ Bridgeport Commu...
- ☒ Bridgewater Medic...
- ☒ Bronx Community C...
- ☒ Brooklyn Heights Cl...
- ☒ Buffalo Health Part...
- ☒ Camden Health Par...
- ☒ Danbury Medical Gr...
- ☒ Elizabeth Care Group
- ☒ Greenwich Medical ...
- ☒ Harrisburg Commu...
- ☒ Hartford Health Ce...
- ☒ Hoboken Wellness ...
- ☒ Jersey City Medical...
- ☒ Lancaster Medical ...
- ☒ Long Island Medical...
- ☒ Manhattan Medical...

Appointment Type

- ☒ (All)
- ☒ Annual Physical
- ☒ Diagnostic Test
- ☒ Follow-Up
- ☒ New Patient
- ☒ Specialist Consulta...
- ☒ Urgent Visit

Specialty

- ☒ (All)
- ☒ Cardiology
- ☒ Dermatology
- ☒ Diagnostic Imaging
- ☒ Neurology
- ☒ Orthopedics
- ☒ Pediatrics
- ☒ Primary Care
- ☒ Urgent Care

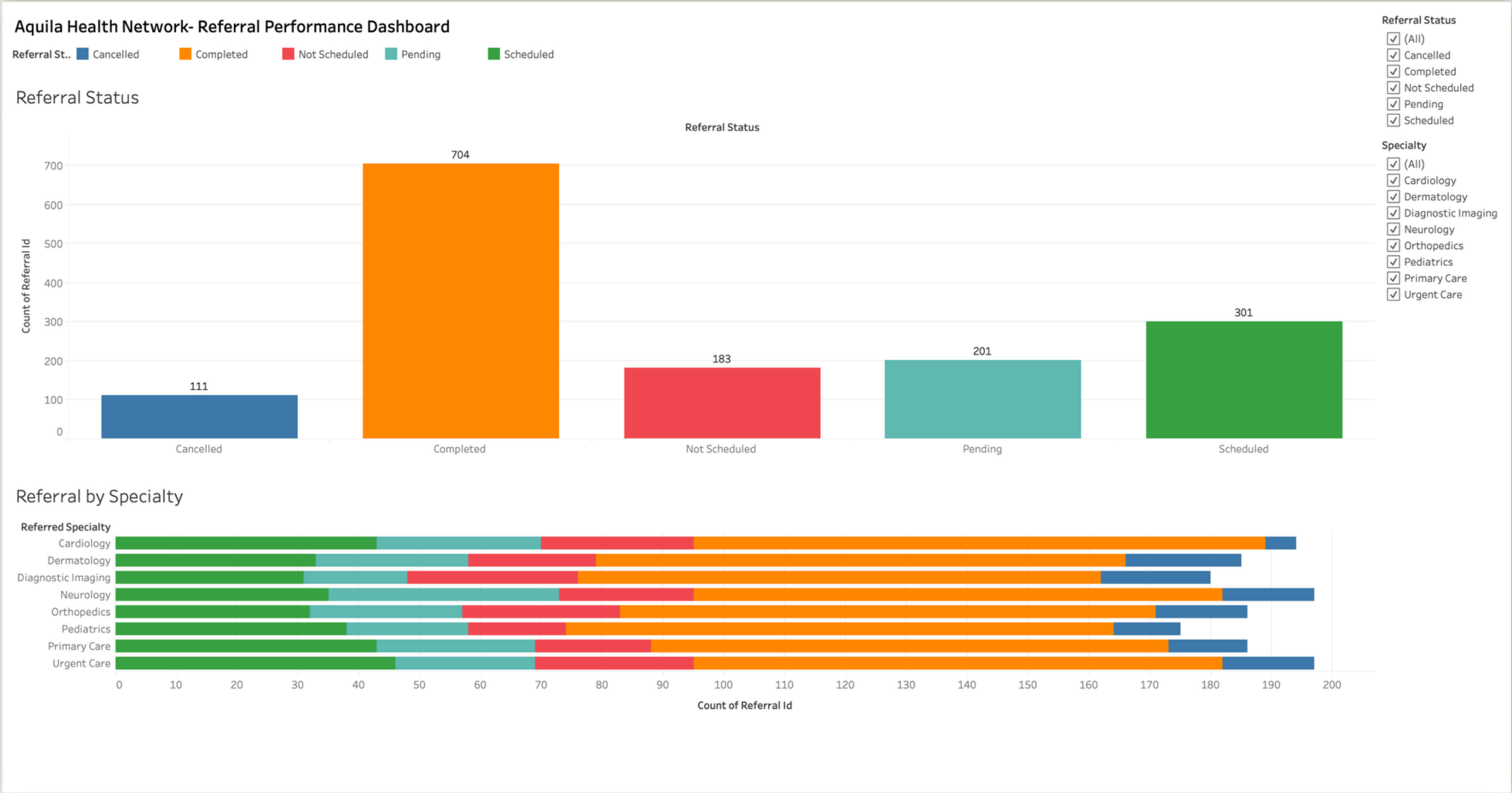
Year

20252026

Provider Utilization Rate

51.1%58.5%

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Dashboard Results: No-Show Risk Dashboard

No-Show Risk Dashboard

935 appointments scored by predicted no-show risk:
187 High risk | 522 Medium risk | 226 Low risk.
Appointments ranked by risk score from highest (0.41)
to lowest, color-coded by risk level for immediate visual
prioritization.

Feature Importance Finding

Prior no-show history (coefficient 0.25) is the dominant
predictor – more than 6x stronger than any other
feature. Appointment type (0.04) and lead time (0.004)
are secondary factors. Day of week and appointment
hour show negative coefficients, reducing risk for
certain scheduling patterns.

Predictive Model Results

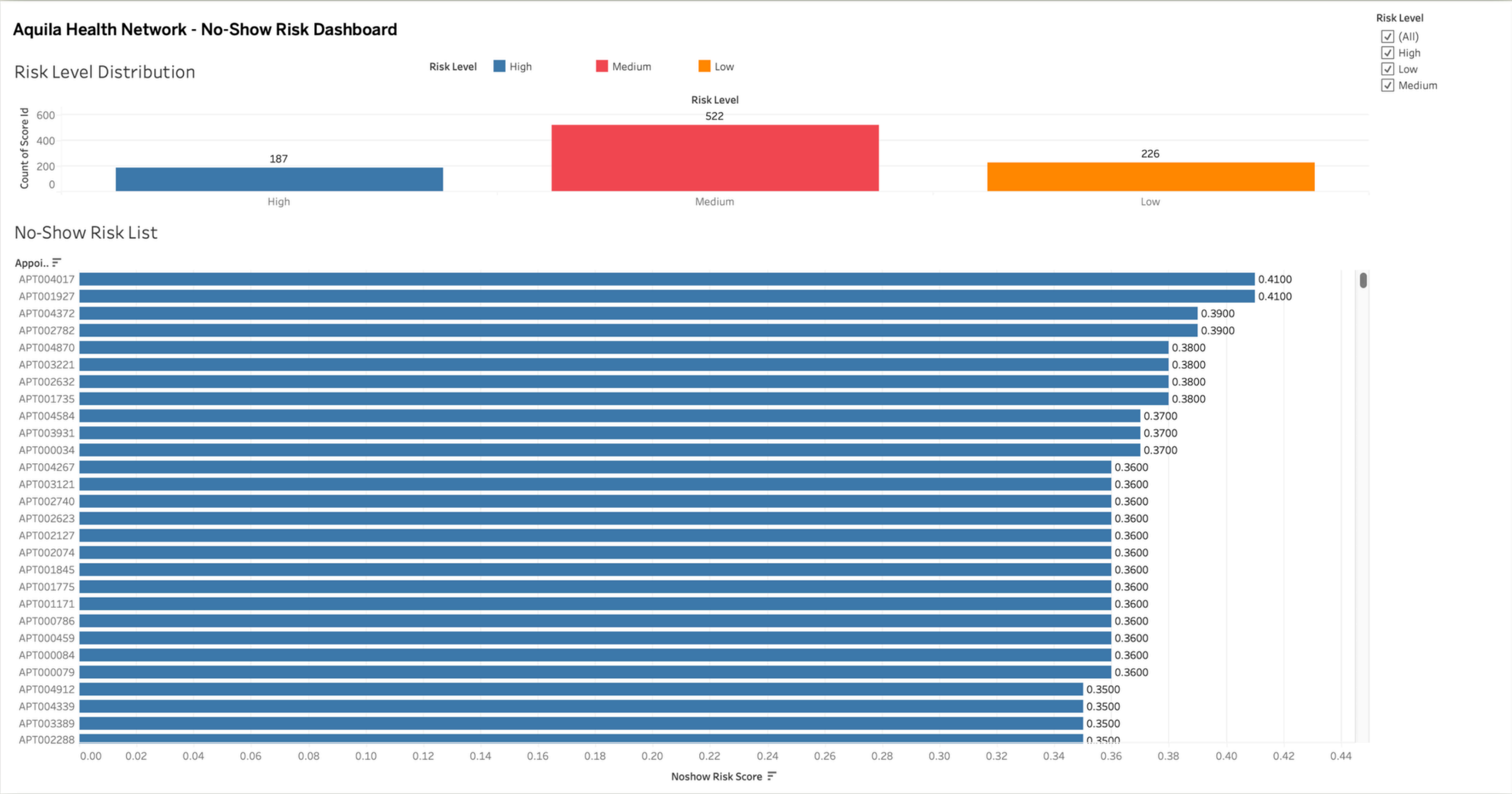
Algorithm: Logistic Regression with balanced class
weighting | ROC-AUC: 0.595 | Training: 4,000 records |
Test: 1,000 records. Model correctly identified 89 of 181
no-show cases in the test set.

Operational Use

Clinic managers can filter the risk list by clinic or date
range and use it to prioritize outreach calls for the
highest-risk upcoming appointments.



Optimizing Clinical Operations Through Enterprise Business Analytics at Aquila Health Network



Aquila Health Network — No-Show Risk Predictive Model

Objective 4: Predictive Analytics Proof of Concept

Author: Karen Lin

Course: MASY GC-4100 Applied Project Capstone

Sponsor: Dan Stone, Damark Group

Overview

This notebook develops a logistic regression model to predict appointment no-show risk at Aquila Health Network. The model scores upcoming appointments by no-show probability using patient history, appointment characteristics, and scheduling patterns.

Model Features

- Lead time (days between booking and appointment)
- Prior no-show count (patient history)
- Appointment hour and day of week
- Appointment type and specialty
- Insurance type

```
[1]: # Import libraries
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix, roc_auc_score
from sklearn.preprocessing import LabelEncoder
import warnings
warnings.filterwarnings('ignore')

print("Libraries loaded successfully")

Libraries loaded successfully
```

5. Model Evaluation

```
[6]: # Generate predictions
y_pred = model.predict(X_test)
y_prob = model.predict_proba(X_test)[:, 1]

# Evaluate
print(f"ROC-AUC Score: {roc_auc_score(y_test, y_prob):.3f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=['Completed', 'No-Show']))
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

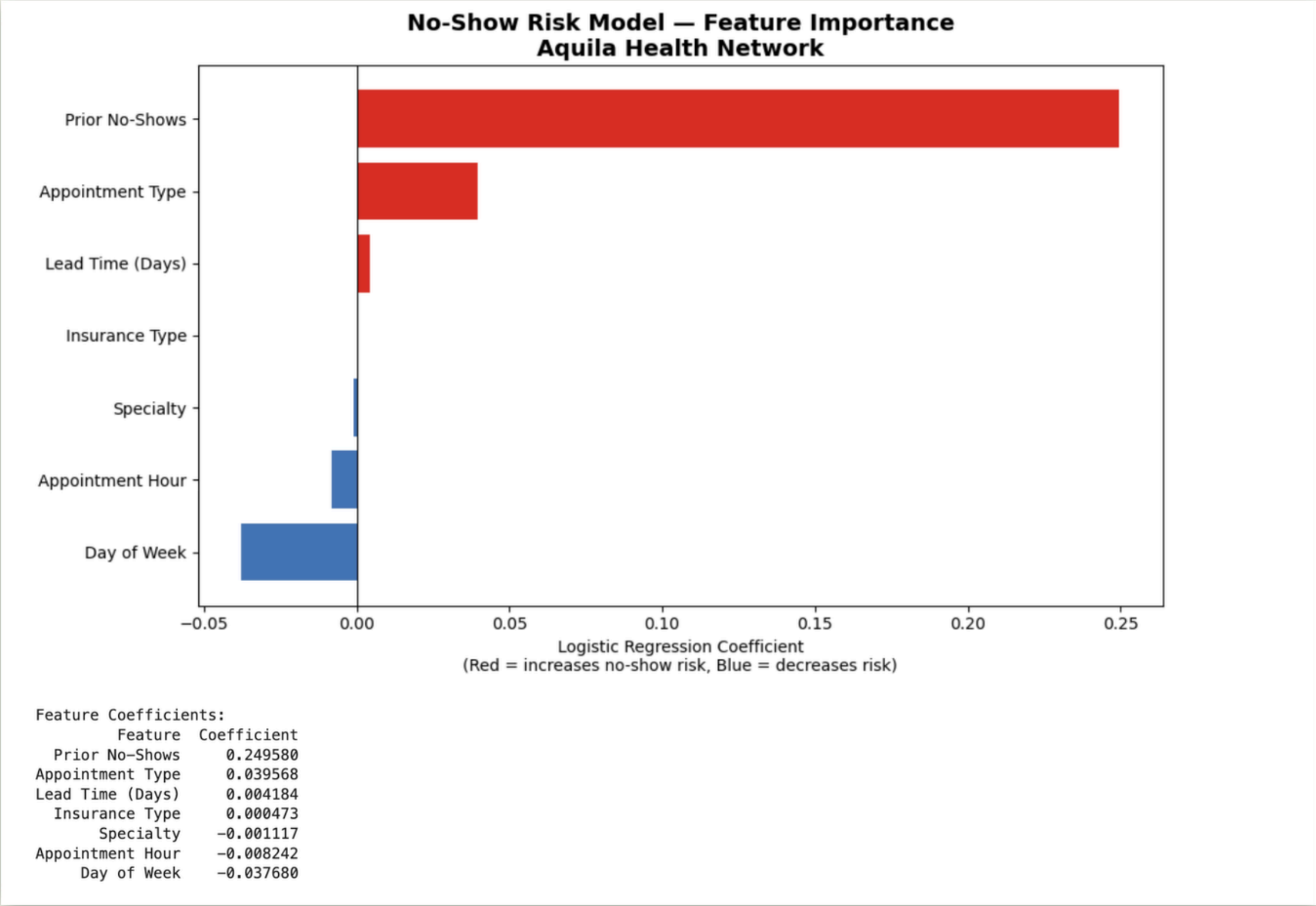
ROC-AUC Score: 0.595

Classification Report:

	precision	recall	f1-score	support
Completed	0.86	0.66	0.75	819
No-Show	0.24	0.49	0.33	181
accuracy			0.63	1000
macro avg	0.55	0.58	0.54	1000
weighted avg	0.74	0.63	0.67	1000

Confusion Matrix:

```
[[543 276]
 [ 92  89]]
```



7. Model Summary and Assumptions

Algorithm: Logistic Regression with balanced class weighting

ROC-AUC Score: 0.595

Key Finding: Prior no-show history is the strongest predictor of future no-show behavior, with a coefficient of 0.25 — more than 6x stronger than any other feature.

Operational Implications:

- Clinic managers should prioritize outreach to patients with prior no-show history
- Friday appointments carry the highest no-show risk (20.9%)
- Wednesday appointments carry the lowest no-show risk (15.8%)

Limitations:

- Model trained on synthetic data — real-world performance may differ
- Logistic regression is a baseline model; more complex models (Random Forest, XGBoost) may improve AUC
- Proof-of-concept only, not intended for production deployment



Key Findings Summary

Finding 1: No-Show Rate at 18.1% – Above Intervention Threshold	Finding 2: No Specialty Meets 82% Utilization Target	Finding 3: 384 Incomplete Referrals Represent Revenue Leakage	Finding 4: Prior No-Show History is the Dominant Predictor	Finding 5: Paterson Community Clinic Has Longest Wait at 26.15 Days
Network no-show rate exceeds the 15% reduction target, with 904 missed appointments out of 5,000. Friday is highest risk day (20.9%), Wednesday lowest (15.8%).	All specialties average between 51–59% utilization, significantly below the 82% target. Network-wide underutilization is driven by no-shows and poor slot recovery.	26% of referrals (384 of 1,500) are pending or not scheduled, representing missed patient care and lost revenue across all specialties.	Patient behavior (coefficient 0.25) is more than 6x stronger than any scheduling factor. Intervention strategy should be patient-centric, not scheduling-centric.	Network average wait is 23.79 days. FRS target is 20% reduction to approximately 19 days. Capacity reallocation from underutilized providers is the primary lever.



Conclusions and Recommendations

Project Success

All 5 objectives completed on schedule. Delivered a fully functional analytics platform with 5 interactive dashboards, a validated star schema data mart, and a proof-of-concept predictive model – all within the 300-hour budget.

Recommendations for Aquila Health Network

1. Implement targeted outreach for patients with prior no-show history – the highest-ROI intervention available.
2. Apply overbooking strategies for high-risk appointment slots (Fridays, long lead times) based on predicted no-show probability.
3. Implement a referral follow-up workflow – any referral Pending or Not Scheduled after 14 days triggers an alert to the referring clinic.
4. Redistribute scheduling capacity from underutilized providers to high-wait clinics to reduce the 23.79-day average wait time toward the 19-day target.

Next Steps

- Integrate platform with live Aquila EHR data
- Upgrade predictive model to Random Forest for improved accuracy
- Expand analytics to inpatient and financial domains
- Deploy dashboards via Tableau Server for organization-wide access.

Thank you.

Karen Lin

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